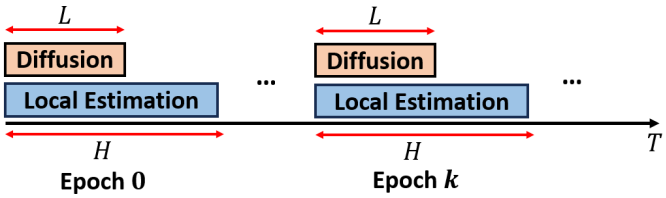
Variance-Reduced Q-Learning over Static and Time-Varying Networks

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Problem and method. N agents learn the optimal Q-function of a shared discounted MDP while communicating over an undirected graph. Each collects T independent generative samples *per state-action pair*. **Variance-Reduced Diffused Q-Learning (VRDQ)** uses H-sample epochs, one Q-table update per epoch, and parallel local operator estimation and consensus diffusion.



Estimation and diffusion run in parallel.

Assumptions and notation. Let $S = |\mathcal{S}|$, $A = |\mathcal{A}|$, $\gamma \in (0,1)$, and $|R(s,a)| \le \bar{R}$, with deterministic rewards. Generative samples are independent across agents and queries. Write $T = KH$, initialize $Q_{i,0} = 0$, and let $e_{i,k} := \|Q_{i,k} - Q^\star\|_\infty$ and $J := \mathbf{1}\mathbf{1}^\top/N$. For a static graph, W is nonnegative, symmetric, doubly stochastic, and primitive, with $\|[W^\ell]_i - \mathbf{1}^\top/N\|_2 \le C_1\rho^\ell$ for all $i \in [N]$, $\ell \ge 0$, and $\rho \in (0,1)$.

Epoch update. Estimate the empirical Bellman optimality operator $\mathcal{T}_{i,k}$ from H fresh samples per pair. In parallel, run L consensus steps on the previous-batch operator evaluations, with $\mathcal{T}_{i,-1} = 0$:

$$d_{i,k}^{(L)} = \sum_{j=1}^{N} (W^L)_{ij} \mathcal{T}_{j,k-1} Q_{j,k}, \qquad Q_{i,k+1} = (1-\alpha) Q_{i,k} + \alpha d_{i,k}^{(L)}.$$

1. Finite-time convergence on static networks. Fix $\delta \in (0,1)$ and choose suitable constants c_1, c_2:

$$K = \left\lceil \frac{c_1 \log(NT)}{1-\gamma} \right\rceil, \qquad \alpha = \frac{\log(NT)}{(1-\gamma)K}, \qquad L = \left\lceil \frac{\log(c_2 N^{3/2}\sqrt{T(1-\gamma)})}{\log(1/\rho)} \right\rceil, \qquad L \le H.$$

With probability at least $1-\delta$, every agent satisfies

$$e_{i,K} \le \frac{e_{i,0}}{NT} + O\left(\frac{\bar{R}\sqrt{\log(NT)\log(2SAT/\delta)}}{(1-\gamma)^{5/2}\sqrt{NT}} \right), \qquad i \in [N]. \tag{1}$$

The $\widetilde{O}(1/\sqrt{NT})$ rate yields an N-fold improvement in per-agent sample complexity, for fixed other parameters.

2. Communication cost and network dependence. Each agent performs $KL = O(\log^2(NT))$ consensus rounds for fixed γ, ρ; each round exchanges an $S \times A$ table with neighbors. Network mixing sets L and the feasibility condition $T \ge KL$, while the statistical error term in (1) has no explicit dependence on topology.

3. Extension to time-varying networks. Let $W(t)$ be nonnegative and doubly stochastic. Suppose that for some integer $B \ge 1$,

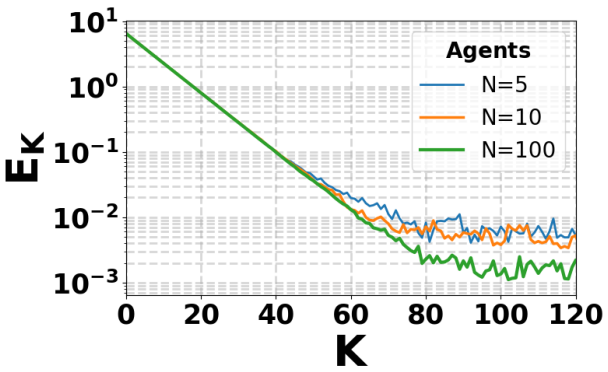
$$\omega := \sup_{t \ge B-1} \|W(t)W(t-1)\cdots W(t-B+1) - J\|_2 < 1.$$

Use $W(kH+\ell)$ during diffusion and choose

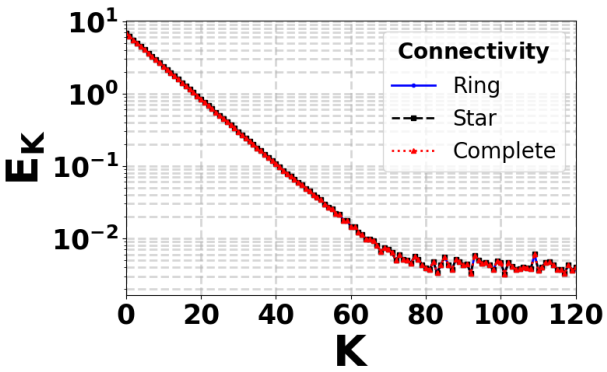
$$\bar{L} = \left\lceil \frac{\log(CN\sqrt{T(1-\gamma)})}{\log(1/\omega)} \right\rceil, \qquad B\bar{L} \le H, \qquad \text{rounds per agent} = KB\bar{L}. \tag{2}$$

The same guarantee (1) holds. Communication is polylogarithmic in T for fixed network parameters; B may depend on N.

Numerical results. $S = 10$, $A = 5$, $\gamma = 0.9$, $R \in [0,1]$, $\alpha = 0.1$, $T = 10^5$, and $\delta = 0.01$. Plots show the average agent error $E_k = N^{-1}\sum_{i=1}^{N} \|Q_{i,k} - Q^\star\|_\infty$.



(a) More agents lower the terminal error.



(b) Similar accuracy across topologies, $N = 100$.